

KOLKATA BOSE INTL, India

WMO: 428090

Lat: 22.655N

Lon: 88.447E

Elev: 16

StdP: 14.69

Time Zone: 5.50 (IND)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	52.1	54.0	44.4	43.2	72.0	46.6	47.0	69.5	12.0	72.1	11.1	71.8	1.4	10	0.336

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	17.1	100.2	81.2	98.3	81.5	96.6	81.0	85.5	94.5	84.8	93.3	84.1	92.1	7.7	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
83.8	176.8	90.5	82.7	170.5	89.4	82.3	168.3	89.0	50.2	94.7	49.3	93.5	48.4	92.2	91.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.3	13.2	11.6	DB	48.9	104.5	1.7	2.6	47.7	106.4	46.7	107.9	45.7	109.4	44.4	111.3	(4)
(5)			WB	47.4	87.6	1.4	1.4	46.4	88.6	45.5	89.4	44.7	90.1	43.7	91.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	80.5	66.3	73.5	81.8	86.8	87.8	86.9	85.2	85.1	85.0	82.5	76.3	68.5	(6)	
	DBStd	7.94	3.62	4.62	4.32	3.80	3.54	3.19	2.41	2.28	2.53	3.07	3.38	3.64	(7)	
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD65	38	27	1	0	0	0	0	0	0	0	0	0	0	(9)	
	CDD50	11133	506	659	985	1103	1170	1107	1092	1087	1050	1009	790	575	(10)	
	CDD65	5695	67	240	520	653	705	657	627	622	600	544	340	120	(11)	
	CDH74	67406	450	1973	5811	8628	9797	8748	7787	7651	7236	5945	2671	709	(12)	
	CDH80	31461	45	659	2769	4761	5624	4533	3407	3281	3085	2384	835	78	(13)	
(14) Wind	WSAvg	4.2	2.7	2.9	4.3	6.7	6.9	5.8	5.2	4.6	4.0	2.8	2.2	2.4	(14)	
(15) Precipitation	PrecAvg	65.9	0.5	0.9	1.1	2.3	5.2	11.6	13.2	13.3	11.3	5.0	1.1	0.3	(15)	
	PrecMax	98.9	5.0	3.4	5.0	8.1	13.7	41.0	26.3	24.4	30.0	14.2	11.7	4.0	(16)	
	PrecMin	42.6	0.0	0.0	0.0	0.0	0.6	3.1	3.5	5.0	1.5	0.0	0.0	0.0	(17)	
	PrecStd	13.5	0.9	0.9	1.2	2.0	3.0	7.1	5.5	5.1	6.4	3.2	2.2	0.8	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	84.1	93.4	98.8	103.6	102.8	100.8	96.6	95.1	95.4	94.8	89.9	84.4	(19)	
		MCWB	68.2	70.4	73.8	77.4	82.1	84.6	83.7	82.8	81.5	80.1	74.2	69.0	(20)	
	2%	DB	80.8	89.6	95.4	100.2	100.0	98.4	93.8	93.3	93.5	92.9	88.1	82.2	(21)	
		MCWB	65.7	70.3	74.1	79.1	83.1	84.3	82.9	82.6	81.8	80.0	73.4	67.8	(22)	
	5%	DB	78.8	86.5	93.6	98.2	98.2	96.4	91.8	91.6	92.2	91.2	86.3	80.5	(23)	
		MCWB	64.0	68.9	73.9	79.6	82.7	83.8	82.4	82.2	81.6	79.2	72.4	66.6	(24)	
	10%	DB	76.7	84.2	91.6	95.4	96.4	93.9	90.0	89.8	91.0	89.5	84.6	78.5	(25)	
		MCWB	63.4	68.0	73.5	79.4	82.2	83.1	81.9	81.8	81.3	78.0	71.4	65.4	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	72.0	76.3	81.3	85.7	87.0	86.9	85.4	85.2	84.6	83.6	78.9	72.6	(27)	
		MCDB	79.5	84.4	91.1	96.6	96.9	96.4	92.5	91.3	91.5	89.9	84.4	79.1	(28)	
	2%	WB	68.9	74.5	79.8	84.1	85.4	85.5	84.3	83.9	83.6	82.3	77.3	70.2	(29)	
		MCDB	76.2	82.1	88.8	95.0	95.3	94.4	90.9	89.8	90.0	88.3	83.3	78.1	(30)	
	5%	WB	66.8	73.0	78.7	82.7	84.6	84.7	83.5	83.2	82.7	81.2	75.9	68.8	(31)	
		MCDB	74.7	80.5	87.0	92.6	94.3	93.1	89.4	88.7	88.6	87.1	82.1	77.1	(32)	
	10%	WB	65.0	71.2	77.5	81.5	83.6	83.9	82.7	82.5	81.9	80.3	74.5	67.2	(33)	
		MCDB	73.2	79.5	85.2	90.6	92.6	91.5	87.9	87.5	87.4	85.9	81.1	75.4	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	20.9	21.1	19.9	18.4	17.1	12.8	10.2	10.0	11.0	13.4	17.6	20.0	(35)	
		MCDBR	23.1	22.8	20.8	19.9	18.3	16.5	12.7	12.2	13.3	15.0	18.0	20.8	(36)	
	5% WB	MCWBR	9.7	8.0	7.9	7.7	7.4	7.2	5.6	5.0	5.1	5.7	6.5	8.6	(37)	
		MCWBR	20.0	19.2	18.1	17.9	17.3	14.7	11.5	11.0	12.0	12.9	14.6	18.8	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.601	0.564	0.602	0.642	0.690	0.689	0.595	0.565	0.573	0.595	0.635	0.646	(40)	
		taud	1.709	1.822	1.784	1.758	1.702	1.714	1.924	2.019	1.980	1.872	1.711	1.667	(41)	
	Ebn at Noon		208	232	230	224	212	211	232	239	234	221	199	190	(42)	
		Edh at Noon	69	65	70	73	77	75	61	56	57	61	68	69	(43)	
(44) All-Sky Solar Radiation	RadAvg		1192	1498	1748	1904	1811	1414	1286	1320	1353	1351	1266	1109	(44)	
(45)	RadStd		73	86	80	93	116	112	104	79	71	102	69	96	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	+0.81	N/A	N/A	N/A	N/A	N/A	+169
(47) Regional (1 neighbor)	N/A	N/A	N/A	+0.81	N/A	N/A	N/A	N/A	N/A	+169

Nomenclature: See separate page